

SYNAPS: A Neuro-Symbolic Framework for Proactive Security in Consumer Electronics

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Abstract—Proactive security risk assessment and prediction (SRAP) is essential to defend against dynamic and evolving threats in the modern interconnected consumer electronics domain. With the rapid emergence of polymorphic attacks and complex network behaviors, traditional systems are often insufficient due to their static nature and limited adaptability. Neural systems alone fail to incorporate the domain expertise necessary for identifying well-known risk indicators in high-stakes cyber-security contexts. To address these gaps, we propose SYNAPS: a neuro-symbolic framework for a proactive security model that fuses data-driven anomaly detection with transparent symbolic reasoning to boost risk scoring and prediction accuracy for consumer device security. Our design first trains an LSTM autoencoder on benign device telemetry to compute continuous neural anomaly scores, and a curated set of device-specific symbolic rules generates interpretable risk scores based on established security indicators. These scores are then blended into a hybrid risk estimate, which feeds into a shallow MLP for multiclass threat classification (NS-MLP). To further strengthen SYNAPS against device-targeted evasions, we introduce three variants: Gaussian-noise augmentation to mimic polymorphic attacks (NS-Aug), adversarial training via a lightweight Generative Adversarial Network (GAN) to model firmware-level exploits (NS-Adv), and transfer learning to adapt our autoencoder to new device models and firmware versions (NS-Trans). Our experiment results indicate that SYNAPS and its enhanced versions substantially outperform neural-only baselines. NS-Adv achieves 93.3% test accuracy with a 12.3% relative improvement over the neural-only model. Overall, the integration of data-driven learning and symbolic reasoning significantly boosts the accuracy and reliability of threat detection in consumer electronics.

Index Terms—Consumer electronics security, neural-symbolic reasoning, symbolic risk optimization, multiclass threat classification, transfer learning, zero-day resilience.

I. INTRODUCTION

CYBER threats in consumer electronic devices are becoming increasingly advanced, adaptive, and stealthy [1]. Attackers now use AI tools to probe firmware vulnerabilities, automate exploitation of smart-home hubs and employ zero-day exploits or compromised insider credentials to persist undetected on devices [2]. Meanwhile, the proliferation of smart wearable health monitors, voice-activated assistants, and connected appliances has expanded the consumer electronics attack surface [3]. Each new device and its accompanying cloud or mobile companion app provides fresh entry points

for adversaries. On the other hand, defenders must grapple with monitoring and securing a heterogeneous ecosystem of resource-constrained endpoints [4]. This rapid evolution underscores the need to move beyond reactive and signature-based defenses toward proactive risk assessment and intelligent threat prediction in consumer electronic environments.

Traditional security mechanisms for consumer electronic devices typically operate reactively [5]. While these defenses can block well-understood malware or repeat attacks, they struggle against novel device-specific threats, such as zero-day firmware vulnerabilities or obfuscated payloads that do not match predefined signatures [6]. To overcome these limitations in the consumer electronics domain, machine learning (ML) techniques have gained traction for automated threat detection and anomaly classification [7]. Meanwhile, deep neural networks can learn from large volumes of device telemetry ranging from sensor readings and API call sequences to network traffic and user interaction patterns to identify subtle deviations indicative of emerging threats [8]. But, recent studies have further found that consumer devices are uniquely vulnerable due to their resource-constrained hardware and diverse vendor-specific firmware [9]. High-profile incidents, such as large-scale IoT botnet attacks and exploitation of insecure smart-home hubs, highlight the urgent need for proactive defenses [10].

To fill this gap, we introduce SYNAPS, a symbolic-neural framework [11] for proactive risk scoring and threat prediction on consumer electronics devices. SYNAPS fuses a single-step LSTM autoencoder trained on benign telemetry (API calls, sensor readings, network flows), which generates continuous neural anomaly scores, with an expert-driven rule engine that produces symbolic risk signals. These streams are combined into a hybrid risk estimate and, together with select raw telemetry features, fed into a shallow MLP classifier (NS-MLP) for multiclass threat detection [12]. The standout advantage of SYNAPS is that it delivers high-accuracy, real-time threat detection on resource-constrained devices. Overall, the contributions of this paper are as follows:

- We design a trustworthy neuro-symbolic framework that combines temporal deep learning with interpretable rule-based reasoning for on-device threat detection and risk scoring.
- We develop a hybrid scoring mechanism that combines the reconstruction error from unsupervised LSTM autoencoders with symbolic scores derived from expert-crafted rules to ensure cross-device consistency.

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- We improve model robustness through three targeted techniques: (i) NS-Aug: Gaussian noise injection to enhance generalization; (ii) NS-Adv: GAN-based generation of adversarial telemetry traces for resilient learning; and (iii) NS-Trans: transfer learning to adapt to firmware environments.
- We present an extensive empirical evaluation of SYNAPS and its variants on a multi-device telemetry dataset.
- We analyze the latency and memory usage of all variants on Mac-based hardware for real-time resource-constrained environments.

The rest of this paper is organized as follows. In Section II, we review prior work on anomaly detection, symbolic reasoning, and neuro-symbolic security systems. Section III introduces our SYNAPS architecture in detail. In Section IV, we describe our experimental setup, datasets, and evaluation metrics, then present and analyze performance results across all SYNAPS variants. Finally, Section V concludes with a summary of findings and outlines future enhancements.

II. RELATED WORKS

Recent advances in cybersecurity have sparked growing interest in neuro-symbolic AI, as detailed in Table I, which seeks to integrate neural networks with the reasoning and interpretability of symbolic systems [13]. This approach is increasingly recognized as a promising solution to address the limitations of purely neural or symbolic models in complex threat areas.

To improve network intrusion detection systems (NIDS), Almadhor et al. [13] introduced a framework that fuses deep learning with symbolic reasoning. Their system achieved better detection accuracy and maintained human-interpretable outcomes. Kalutharage et al. [14] explored the use of reinforcement learning in combination with symbolic reasoning for cyber defense. Their system was designed to generate timely and explainable responses to evolving threats using symbolic knowledge to guide the learning process. In the domain of explainable AI for privacy and threat modeling, Piplai et al. [15] proposed a neuro-symbolic framework that integrates knowledge graphs with neural learning. Their work addresses the crucial challenge of explainability in cybersecurity using structured domain knowledge. Similarly, Himmelhuber et al. [16] demonstrated how sub-symbolic models could be enhanced through symbolic overlays. Their approach focuses on the explanation of cyberattacks for a more intuitive and traceable detection process.

In the context of cloud-based intrusion detection, a recent study [22] applied neuro-symbolic methods to enable real-time threat detection and monitoring. The model proved effective in cloud-native environments. Bertrand Van Ouytsel et al. [17] introduced an approach combining symbolic system call analysis with federated learning. It enables collaborative malware identification without sharing raw data. However, Jagatheesaperumal et al. [18] proposed a neuro-symbolic perception framework to enhance situational awareness and resilience. Deep perceptual models with formal symbolic constraints ensure safe and reliable operation in adversarial environments. In efforts to support real-time cyber threat

Algorithm 1 Neural Anomaly Detection

Input : Time-series dataset $X \in \mathbb{R}^{T_{\max} \times d}$, window length T , false-alarm rate α_{FA}

Output: Reconstruction-error threshold τ , anomaly scores $S_{\text{anom}}[t] \forall t$

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1 Initialize LSTM-AE parameters  $\theta$ ;
2 for  $epoch \leftarrow 1$  to  $N_{\text{epochs}}$  do
3   Sample minibatch of windows  $\{\mathbf{x}_{t-T+1:t}\}$  from
   nominal  $X$ ;
4   Compute reconstruction  $\hat{\mathbf{x}}_{t-T+1:t} = f_{\theta}(\mathbf{x}_{t-T+1:t})$ ;
5   Update  $\theta$  to minimize  $\|\mathbf{x} - \hat{\mathbf{x}}\|^2$ ;
6 Initialize list  $\mathcal{E}_{\text{train}} \leftarrow []$ ;
7 for  $t \leftarrow T, \dots, T_{\max}$  do
8    $\hat{\mathbf{x}}_t \leftarrow f_{\theta}(\mathbf{x}_{t-T+1:t})$ ;
9    $e_t \leftarrow \|\mathbf{x}_t - \hat{\mathbf{x}}_t\|_2^2$ ;
10  append  $e_t$  to  $\mathcal{E}_{\text{train}}$ ;
11  $\tau \leftarrow$  the  $(1 - \alpha_{\text{FA}})$  quantile of  $\mathcal{E}_{\text{train}}$ ;
12 for  $t \leftarrow T, \dots, T_{\max}$  do
13    $\hat{\mathbf{x}}_t \leftarrow f_{\theta}(\mathbf{x}_{t-T+1:t})$ ;
14    $S_{\text{anom}}[t] \leftarrow \|\mathbf{x}_t - \hat{\mathbf{x}}_t\|_2^2$ ;
15 return  $\tau, S_{\text{anom}}$ 

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detection, another study [14] presented a system that merges deep anomaly detection with rule-based symbolic reasoning. This work balances data-driven adaptability with transparent rule logic. For proactive threat modeling, Lei et al. [19] developed the ADAPT framework, which integrates game-theoretic reasoning with neuro-symbolic agents.

Finally, a recent study in [23] explored the role of neuro-symbolic AI in incident response and analyst decision support. Through initial case studies, the authors demonstrated how symbolic knowledge can be embedded into neural pipelines to bridge the gap between theoretical AI capabilities and operational cybersecurity needs. Many existing approaches focus on narrow applications such as static intrusion detection or policy enforcement, with limited adaptability to evolving threats and polymorphic attack patterns. These gaps motivate the development of our proposed system, which aims to deliver a robust approach to threat detection in consumer electronics security systems.

III. PROPOSED METHODOLOGY

The proposed SYNAPS is designed with a dual-pipeline neuro-symbolic architecture that unifies data-driven learning with expert-defined reasoning for proactive cybersecurity. The system processes incoming device telemetry through two parallel paths: (1) a neural anomaly detection pipeline, where an LSTM autoencoder models normal behavioral patterns and produces a continuous anomaly score, and (2) a symbolic reasoning pipeline, where a curated set of human-defined security rules computes risk indicators based on policy violations. These two risk signals are fused into a hybrid risk score that balances contextual adaptability with explainable logic.

TABLE I
COMPARATIVE SUMMARY OF NEURO-SYMBOLIC CYBERSECURITY APPROACHES

Ref./Author	Application Area	Neural Component	Symbolic Component	Advantage	Limitation
Almadhor et al. [13]	Network Intrusion Detection	Deep Neural Networks	Rule-Based Reasoning	Enhanced detection accuracy and interpretability	Tested primarily on known attacks
Kalutharage et al. [14]	Cyber Defense Automation	Reinforcement Learning	Symbolic Planning/Reasoning	Explainable, adaptive threat response	Complex integration pipeline
Piplai et al. [15]	Privacy and Explainable AI	Knowledge-Graph Enhanced Networks	Knowledge Graphs	High explainability and contextual learning	Scalability of symbolic inference
Himmelhuber et al. [16]	Attack Detection and Explanation	Graph Neural Networks	Attack Rule Filters, Ontologies	Better false-positive reduction	Relies on handcrafted rules
Bertrand et al. [17]	Malware Detection in Federated Learning	Federated ML Models	System Call Dependency Graphs	Privacy-preserving malware detection	Depends on quality of symbolic abstraction
Jagatheesaperum et al. [18]	Cyber-Physical Systems Security	Perception DNNs	Safety Constraints and Formal Logic	Resilience in safety-critical environments	Evaluation focused on simulation
Lei et al. [19]	Penetration Testing Automation	Q-learning Agents	Game-Theoretic Rules	Proactive vulnerability modeling	Simulation-heavy, needs real validation
Abdallah et al. [20]	Cloud Intrusion Detection	DNNs for Anomaly Detection	Policy Compliance Rules	Real-time detection and compliance	Lacks deployment benchmarks
Jablaoui et al. [21]	SOC Incident Response	CNN/RNN + Decision Layers	Incident Ontologies	Bridges theory and real-world use	Early-stage case studies
Proposed (SYNAPS)	Risk Scoring and Threat Detection	LSTM Autoencoder + MLP (NS-MLP)	Expert-Defined Symbolic Rules	High accuracy and robustness via variants (NS-Adv, NS-Aug, NS-Trans)	Requires careful rule design and tuning of hybrid fusion

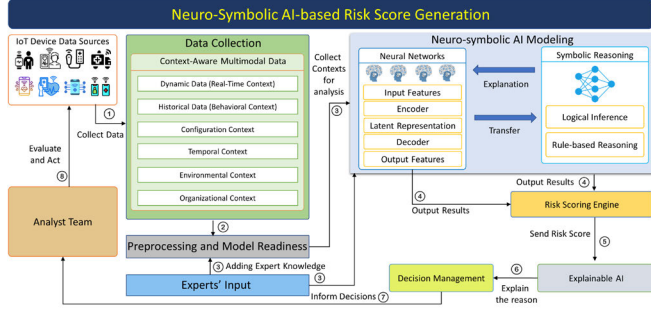


Fig. 1. The risk assessment module in SYNAPS.

A. Risk Score Generation

This component of the SYNAPS is depicted in Fig. 1, which provides context-aware cybersecurity risk assessments. The neuro-symbolic (NS) design fuses data-driven learning with expert-defined rules. The neural network captures nonlinear patterns and contextual anomalies from device telemetry, and the symbolic reasoning layer enforces domain-specific policies derived from expert knowledge.

1) *Model Initialization*: First, we define the input data as a sequence of feature vectors: $X = \{x_1, x_2, \dots, x_n\}$, $x_i \in \mathbb{R}^d$ where each x_i is a multidimensional feature set extracted from system behavior such as login timestamps, activities, file accesses, and network flows.

2) *Neural Component*: Once the input feature sequences are defined, they are passed into the neural module, which learns to model temporal dependencies and behavioral patterns that simple rule-based systems cannot capture. We employ an LSTM-based autoencoder [24], where the encoder compresses a sliding window of past observations ($x_{i-T:i}$) into a latent vector z_i , and the decoder reconstructs the sequence $\hat{x}_{i-T:i}$:

$$\mathcal{L}_{\text{rec}}(x_{i-T:i}) = \frac{1}{T+1} \sum_{j=i-T}^i \|x_j - \hat{x}_j\|_2^2 \quad (1)$$

After minimizing this loss, the network learns benign operating patterns. The hidden state h_i from the LSTM captures temporal context and is projected into a probabilistic neural anomaly score:

$$h_i = \text{LSTM}(x_{i-T:i}), \quad f_{NN}(x_i) = \sigma(W h_i + b) \quad (2)$$

where W and b are learnable parameters and $\sigma(\cdot)$ is a sigmoid activation mapping the score to $[0, 1]$.

3) *Symbolic Component and Fusion (NS-Only)*: After deriving the neural anomaly scores, the outputs are passed to the symbolic reasoning module, which interprets them alongside predefined security rules to produce an explainable risk estimate. This module $g_{SR}(x_i)$ evaluates expert-defined security rules and computes a normalized symbolic risk score. Each rule r_i has a severity weight w_i normalized such that $\sum_i w_i = 1$. The final hybrid risk estimate integrates neural and symbolic evidence as follows:

$$R(x_i) = \alpha f_{NN}(x_i) + (1 - \alpha) g_{SR}(x_i) \quad (3)$$

where α controls the contribution of the neural component and $(1 - \alpha)$ the symbolic one. The NS-Only model terminates here and uses $R(x_i)$ as a continuous risk level for binary assessment (benign vs. malicious).

4) *Extension to Multiclass Threat Prediction (NS-MLP)*: To move beyond binary decisions, the NS-MLP variant uses the fused NS representation as input to a shallow MLP. The neural hidden state h_i , symbolic activations $g_{SR}(x_i)$, and hybrid risk score $R(x_i)$ are concatenated into:

$$\mathbf{z}_i = [h_i \parallel g_{SR}(x_i) \parallel R(x_i)] \quad (4)$$

The vector $\mathbf{z}_i$ is then passed through a lightweight MLP classifier:

$$\hat{y}_i = \text{Softmax}(W_2 \text{ReLU}(W_1 \mathbf{z}_i + b_1) + b_2) \quad (5)$$

This extension differentiates among multiple attack types to transform fused risk estimates into fine-grained threat labels.

Algorithm 2 Symbolic Risk Weight Optimization

Input : Set of symbolic rules $\{r_i\}$ with base weights w_i and observed violation rates v_i
Output: Updated risk weights $\tilde{w}_i$

- 1 **foreach** rule r_i **do**
- 2 Compute adjusted weight: $\tilde{w}_i = w_i \cdot (1 + \lambda v_i)$;
- 3 **Normalize** weights: $\tilde{w}_i = \tilde{w}_i / \sum_j \tilde{w}_j$;
- 4 **return** $\{\tilde{w}_i\}$

5) *Hyperparameter Selection*: Once the architecture is established, the main hyperparameters are optimized to balance accuracy in the NS-MLP model. To optimize the LSTM window size T , we computed autocorrelations of each time-series feature and conducted a validation grid search $T \in \{5, 10, 20, 30, 50\}$ by selecting $T = 10$ for the best reconstruction-error AUC. In $g_{SR}(\cdot)$, rule weights w_i were fine-tuned by $\pm 10\%$ on validation data to maximize symbolic score. The fusion weight α was optimized over $\alpha \in \{0.0, 0.1, \dots, 1.0\}$, with $\alpha = 0.7$ giving the highest validation accuracy. Finally, the binary decision output is obtained by thresholding $R(x_i)$ at τ :

$$\hat{y}_i = \begin{cases} 1, & \text{if } R(x_i) \geq \tau, \\ 0, & \text{otherwise.} \end{cases} \quad (6)$$

B. Symbolic Logic Layer

After tuning the key hyperparameters, SYNAPS proceeds to the symbolic reasoning layer, which applies explicit rules to derive policy-aligned risk scores. The symbolic layer computes a risk score based on rule violations. Each rule represents a specific security policy or behavioral constraint, such as login limits or access restrictions. Symbolic rules $\{r_j\}_{j=1}^m$ originate from three sources: (i) domain and compliance policies, (ii) incident post-mortems and near-miss reports, and (iii) patterns surfaced by the neural component (LSTM-AE) through reconstruction error and attention on features. A candidate rule r_j is admitted only if it satisfies *coverage* (flags $\geq \theta_{cov}$ fraction of historical incidents) and *specificity* (false-positive rate $\leq \theta_{fpr}$) on a validation set. Each rule is specified as a temporal predicate φ_j over a sliding window W_t :

$$\delta_j(x_i, t) \in \{0, 1\}, \delta_j(x_i, t) = \mathbb{1}[\varphi_j(x_{i, [t-|W_t|+1:t]})] \quad (7)$$

To handle uncertainty, we also define a *soft indicator* $\tilde{\delta}_j(x_i, t) = \sigma((s_j(x_i, t) - \tau_j)/\gamma_j)$ with logistic $\sigma(\cdot)$, score $s_j(\cdot)$, threshold τ_j , and temperature γ_j . Recent violations are emphasized via time decay $d(\Delta) = \exp(-\lambda\Delta)$. For sequence x_i , the LSTM autoencoder produces per-timestep error $e_t = \|x_t - \hat{x}_t\|_2$ and latent embeddings z_t . Rules may depend on (e_t, z_t) , which is linked to learned anomalies and context with explicit constraints:

$$\varphi_{\text{spike}} : \frac{1}{K} \sum_{k=0}^{K-1} \mathbb{1}[e_{t-k} > \tau_e] \geq \rho, \varphi_{\text{context}} : z_t \in \mathcal{C}_{\text{restricted}}, \quad (8)$$

Each rule weight w_j reflects impact and reliability. Let I_j be the operational impact (0–1), estimated with a risk

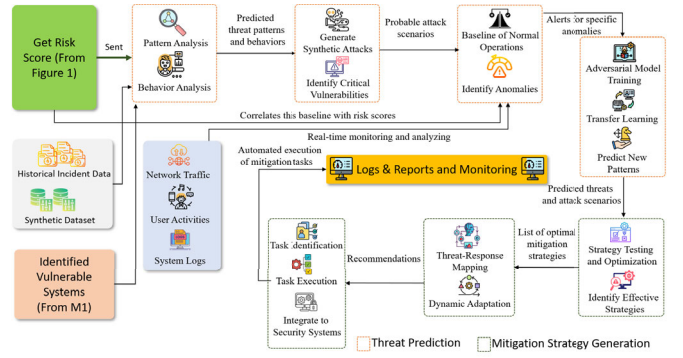


Fig. 2. Overview of the SYNAPS threat prediction pipeline.

matrix score. $\text{TPR}_j, \text{FPR}_j$ be empirical rates on validation and $C_{fp}, C_{fn} > 0$ the costs. We define a utility-aligned weight $w_j \propto I_j \cdot (\alpha \text{TPR}_j - \beta \text{FPR}_j)$, $\alpha = C_{fn}$, $\beta = C_{fp}$ and calibrate the proportionality by $\sum_j w_j = 1$. Thresholds τ_j are selected by maximizing Youden's $J = \text{TPR} - \text{FPR}$ subject to $\text{FPR} \leq \theta_{fpr}$ (operating constraint). To avoid redundant rules, we solve a sparse selection [25] as given below:

$$\min_{\mathbf{w} \geq 0} \mathbb{E}[\ell(y, \sum_j w_j \tilde{\delta}_j)] + \lambda_1 \|\mathbf{w}\|_1 + \lambda_2 \Omega_{\text{overlap}}(\mathbf{w}) \quad (9)$$

where ℓ is logistic loss on labeled incidents $y \in \{0, 1\}$ and Ω_{overlap} penalizes jointly active highly correlated rules. With time decay $d(\Delta)$ the score is:

$$g_{SR}(x_i, t) = \sum_{j=1}^m w_j \sum_{\Delta=0}^{|W_t|-1} d(\Delta) \tilde{\delta}_j(x_i, t - \Delta)$$

$$\tilde{g}_{SR} = \frac{g_{SR}}{\sum_j w_j \sum_{\Delta} d(\Delta)} \in [0, 1]. \quad (10)$$

We evaluate each rule r_j by removing it and measuring the change in AUROC and F1 for incident detection. We also test how sensitive the risk score $\tilde{g}_{SR}$ is to $\pm 10\%$ adjustments in its weights w_j and thresholds τ_j , and we verify that the score is well-calibrated to the observed risk. Each alert also reports which rules were triggered, their individual contributions $w_j \cdot \tilde{\delta}_j$, and the corresponding time window.

C. Threat Prediction Pipeline

To extend beyond basic anomaly detection, SYNAPS incorporates a threat prediction and simulation pipeline, as shown in Fig. 2. It advances the NS model by adding layers for synthetic attack generation, vulnerability prioritization, and adversarial training.

1) *Synthetic Attack Generation (NS-Aug)*: A key limitation of device-level security datasets is the lack of diversity in advanced or stealthy attack samples. To mitigate this, the NS-Aug variant introduces a synthetic attack generation mechanism that expands the training data using domain-aware transformations. Known attack traces x_{real} are perturbed to create realistic variants:

$$x_{\text{synthetic}} = \mathbb{T}(x_{\text{real}}, \theta) \theta \sim \mathcal{U}(\theta_{\min}, \theta_{\max}) \quad (11)$$

where $\mathbb{T}$ is a transformation function (e.g., feature masking) parameterized by θ . These additional synthetic samples diversify the model's training distribution and reduce overfitting for better generalization to unseen attack patterns.

2) *Adversarial Training (NS-Adv)*: While synthetic augmentation enhances diversity, it does not guarantee robustness against deliberately crafted evasions. To counter this, the NS-Adv variant employs adversarial learning using a Generative Adversarial Network (GAN) [26]. The generator G produces perturbed attack examples designed to fool the discriminator D , which learns to distinguish real from synthetic inputs:

$$\min_G \max_D V(D, G) = \mathbb{E}_{x \sim p_{\text{real}}} [\log D(x)] + \mathbb{E}_{z \sim p_z} [\log(1 - D(G(z)))] \quad (12)$$

This min-max training process hardens the classifier by exposing it to evasive attack patterns for improving resilience to zero-day and adversarial scenarios.

3) *Critical Vulnerability Identification*: To focus learning and reasoning on the most severe threats, SYNAPS incorporates a vulnerability prioritization mechanism that adjusts symbolic rule weights based on empirical exploit data. Each vulnerability v_i is scored by its observed frequency impact:

$$\text{Score}_{\text{vuln}}(v_i) = \text{freq}(v_i) \times \text{impact}(v_i) \quad (13)$$

$$\widetilde{\text{Score}}_{\text{vuln}}(v_i) = \frac{\text{Score}_{\text{vuln}}(v_i)}{\max_j \text{Score}_{\text{vuln}}(v_j)} \quad (14)$$

The normalized score $\widetilde{\text{Score}}_{\text{vuln}} \in [0, 1]$ dynamically scales each symbolic rule weight. These normalized priorities then modulate our symbolic rule weights w_j :

$$w_j \leftarrow w_j \times (1 + \widetilde{\text{Score}}_{\text{vuln}}(v_{r_j})) \quad (15)$$

Rules tied to high-risk vulnerabilities have proportionally greater influence in the overall risk computation. Each symbolic rule is represented as a pair: $R_i = (\text{condition}_i, \text{severity}_i)$, where $\text{severity}_i \in \{1, 2, 3\}$. where condition_i defines a security policy and severity_i quantifies its criticality. The aggregated symbolic score for a device instance is then normalized as follows:

$$S_{\text{raw}} = \sum_{i=1}^n a_i \cdot \text{severity}_i \quad (16)$$

where $a_i \in \{0, 1\}$ indicates whether the rule i is triggered (1) or not (0). And, n is the total number of symbolic rules defined for that device class. This score reflects the cumulative symbolic risk before normalization. To map raw symbolic scores, we perform min-max normalization based on the maximum possible symbolic score:

$$S_{\text{sym}} = \frac{S_{\text{raw}}}{S_{\text{max}}}, \quad \text{where } S_{\text{max}} = \sum_{i=1}^n \text{severity}_i \quad (17)$$

4) *Anomaly Detection and Threat Classification (NS-MLP)*: To model normal operational behavior, SYNAPS performs

baseline profiling over a historical window of $\mathcal{T}$ time steps. For each feature, the mean and variance are computed as follows:

$$\mu_t = \frac{1}{\mathcal{T}} \sum_{i=1}^{\mathcal{T}} x_i \quad (18)$$

However, systems naturally exhibit variation, so the framework also calculates the variance σ_t^2 to measure typical fluctuations:

$$\sigma_t^2 = \frac{1}{\mathcal{T}} \sum_{i=1}^{\mathcal{T}} (x_i - \mu_t)^2 \quad (19)$$

and deviations from the baseline are flagged when:

$$|x - \mu_t| > k \cdot \sigma_t \quad (20)$$

Here x is the newly observed value, μ_t is the previously computed baseline mean, and k is a tunable threshold parameter that controls the system's sensitivity. A high threshold $k = 3$ flags only extreme outliers, and smaller values (1.5–2) capture subtle deviations relevant to insider or firmware-level threats. Finally, to ensure model portability across heterogeneous devices and firmware versions, SYNAPS incorporates a cross-device transfer learning module (NS-Trans) described in the following section.

5) *Cross-Device Transfer Learning Module (NS-Trans)*: To enable reliable transfer across devices operating under different firmware versions, NS-Trans relies on a shared telemetry schema rather than identical raw data [27]. All participating devices record standardized behavioral indicators to ensure that each input sample x_i lies within a common feature space $\mathbb{R}^d$, where each dimension corresponds to a semantically consistent measurement. During transfer, the pretrained source model $f_{\text{source}}(x)$ is augmented by a lightweight correction term $\Delta f(x; \theta_{\Delta})$, parameterized by θ_{Δ} and fine-tuned using a small set of labeled target-domain samples $(x_i, y_i)_{i=1}^N$. The objective minimizes a regularized empirical loss to adapt the model with minimal parameter updates, as given below:

$$\theta_{\Delta}^* = \arg \min_{\theta_{\Delta}} \frac{1}{N} \sum_{i=1}^N \mathcal{L}(y_i, f_{\text{source}}(x_i) + \Delta f(x_i; \theta_{\Delta})) + \lambda \|\theta_{\Delta}\|_2^2 \quad (21)$$

Here, $\mathcal{L}$ denotes the loss function and λ controls the strength of $\mathcal{L}$ regularization to limit the correction magnitude. In our implementation, $\lambda = 0.01$ provided stable adaptation and prevented overfitting. The correction layer $\Delta f(x; \theta_{\Delta})$ is realized as a single dense layer stacked over frozen intermediate representations from the LSTM encoder. Once trained, the adapted model is obtained as stated below:

$$f_{\text{target}}(x) = f_{\text{source}}(x) + \Delta f(x; \theta_{\Delta}^*) \quad (22)$$

In practice, the correction function can be expressed as a simple affine transformation:

$$\Delta f(x; \theta_{\Delta}) = W_{\Delta} \phi(x) + b_{\Delta} \quad (23)$$

where $\phi(x)$ is the fixed feature representation extracted from f_{source} , and W_{Δ}, b_{Δ} are learnable adaptation parameters. This strategy enables cross-device generalization with minimal

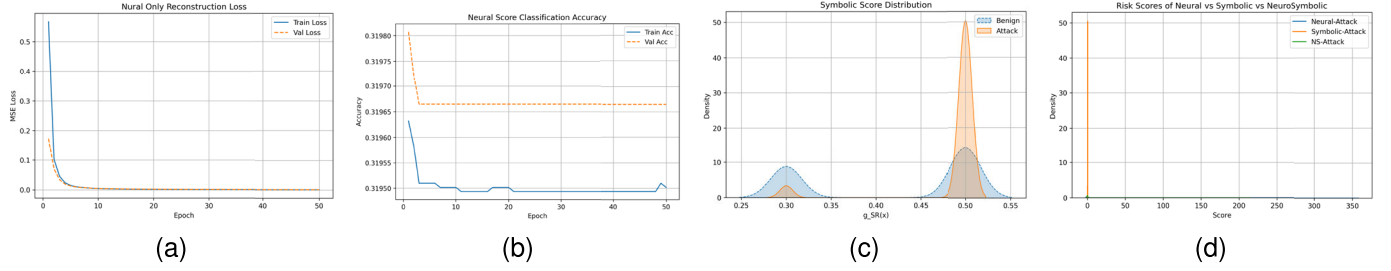


Fig. 3. Model score behaviors observed with a window size $T = 10$, hidden dimension $H = 128$, batch size = 64, and learning rate = 10^{-3} . (a) Reconstruction loss convergence. (b) Neural-score classification accuracy. (c) Distribution of symbolic rule-based scores across varying severity levels. (d) Combined hybrid risk scores $\alpha = 0.7$.

TABLE II
EXPERIMENTAL SETUP AND TOOLS

Component	Specification / Library (Version)
Hardware	Intel Core i9 CPU; 64 GB RAM
Operating System	Ubuntu 22.04 LTS
Deep Learning Framework	TensorFlow 2.11 & Keras
Symbolic Engine	Rule-based logic
Data Handling	Pandas; NumPy
Preprocessing & Splitting	Scikit-learn 1.3
Modeling (Classifiers)	Scikit-learn 1.3
Evaluation Metrics	Scikit-learn 1.3
Visualization	Matplotlib
Dataset	UNSW-NB15 [29]

retraining cost and ensures compatibility across firmware versions, which makes NS-Trans suitable for resource-constrained IoT environments.

IV. EXPERIMENTAL RESULTS AND ANALYSIS

We examine each component of SYNAPS: risk scoring, anomaly detection, and threat prediction by comparing Neural-Only, Symbolic-Only, NS-Only, and NS variants (NS-MLP, NS-Aug, NS-Adv, and NS-Trans).

A. Environment Setup

The experimental setup and tools are summarized in Table II. The symbolic rules were implemented in Python as a custom logic engine. Each experiment is repeated five times with different random seeds, and the mean values of accuracy, precision, recall, F1-score, and ROC-AUC are reported. The experiments are fully reproducible, and performance comparisons across all SYNAPS variants are fair and consistent. Our experiment code is available on the GitHub repository [29].

B. Data Loading and Preprocessing

The UNSW-NB15 dataset [28] is used to evaluate the proposed SYNAPS. The dataset covers ten traffic classes: one normal class and nine attack categories. The testing set contains 82,332 samples, and the training set includes 175,341 samples. Before training, all records are cleaned by removing identifier fields, categorical variables are one-hot encoded, and all numeric features are scaled to the $[0,1]$ range using min-max normalization. The dataset is then divided into 70% training, 20% validation, and 10% test subsets using stratified sampling to preserve class distribution.

C. Model Behavior and Anomaly Differentiation

As illustrated in Fig. 3, the anomaly detection components of SYNAPS effectively separate normal and malicious device behaviors. The LSTM autoencoder quickly learns compact representations of benign telemetry patterns, and the symbolic rule engine provides policy-driven risk signals. When fused, the combined scores produce an amplified separation between benign and attack distributions so that the NS fusion enhances anomaly discrimination and robustness on edge devices.

D. Threat Prediction Metrics Across Models

As part of the experimental evaluation, we benchmarked all SYNAPS variants against pure neural and symbolic baselines to assess predictive robustness and adaptability under realistic IoT conditions. The results (Fig. 4) show that SYNAPS consistently outperforms both standalone components across all key metrics. In particular, NS-Adv achieves the highest accuracy, F1 score, and ROC-AUC, as summarized in Table III. NS variants such as NS-Aug, NS-MLP, and NS-Trans also yield significant gains and enhance threat prediction in resource-constrained environments.

E. Class Imbalance Mitigation

Telemetry-based threat detection often suffers from severe class imbalance [30], where benign samples dominate, and certain attack types appear rarely. To address the class imbalance inherent in telemetry-based threat detection, we apply strategies at both the anomaly detection (LSTM autoencoder) and classification (NS-MLP) stages. On the other hand, the NS-MLP classifier is responsible for multiclass threat prediction that directly addresses class imbalance using a weighted cross-entropy loss function.

F. Symbolic Score Analysis

To demonstrate the discriminative power of the symbolic risk component, we analyzed two representative telemetry logs: one benign and one malicious. In the benign example (Listing 1), the device executed standard API calls (`open_socket`, `read_config`, `send_data`) and accessed only a secure port. The symbolic rule engine matched no rules with a raw symbolic score of $S_{\text{raw}} = 0$, and consequently a normalized symbolic risk score of

TABLE III
PERFORMANCE COMPARISON ACROSS MODELS AND VARIANTS

Model	Training					Validation					Test				
	Accuracy	Precision	Recall	F1	ROC-AUC	Accuracy	Precision	Recall	F1	ROC-AUC	Accuracy	Precision	Recall	F1	ROC-AUC
Neural-Only	0.8292	0.8889	0.8561	0.8722	0.9097	0.8304	0.8916	0.8546	0.8727	0.9086	0.8305	0.8918	0.8545	0.8728	0.9075
Symbolic-Only	0.7571	0.7623	0.9346	0.8397	0.6567	0.7606	0.7639	0.9382	0.8421	0.6601	0.7596	0.7633	0.9374	0.8415	0.6590
NS Only	0.8561	0.8570	0.9466	0.8996	0.9307	0.8542	0.8562	0.9443	0.8981	0.9302	0.8558	0.8585	0.9436	0.8991	0.9282
NS-MLP	0.9298	0.9108	0.9942	0.9507	0.9696	0.9292	0.9101	0.9942	0.9503	0.9686	0.9325	0.9132	0.9956	0.9526	0.9691
NS-Aug	0.9390	0.9386	0.9839	0.9607	0.9775	0.9267	0.9118	0.9880	0.9483	0.9698	0.9307	0.9151	0.9900	0.9511	0.9714
NS-Adv	0.9650	0.9626	0.9970	0.9795	0.9884	0.9297	0.9114	0.9931	0.9505	0.9703	0.9326	0.9149	0.9935	0.9526	0.9710
NS-Trans	0.9252	0.9054	0.9940	0.9476	0.9555	0.9252	0.9048	0.9948	0.9477	0.9541	0.9275	0.9078	0.9945	0.9491	0.9559

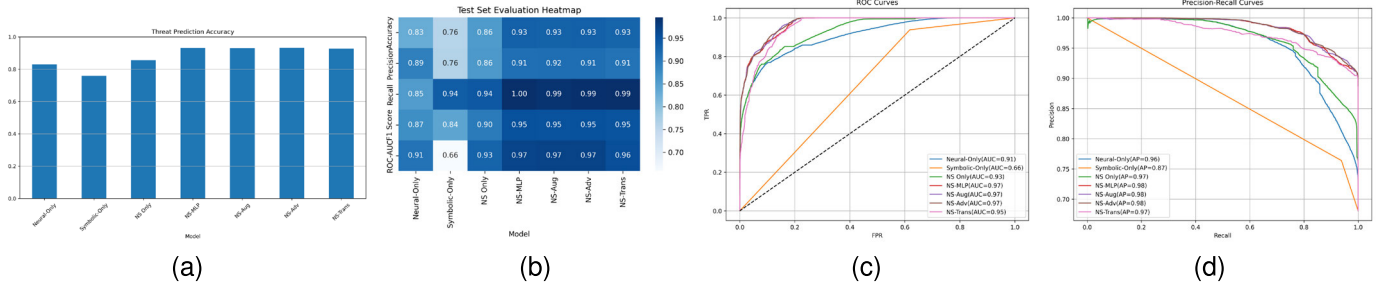


Fig. 4. Test set evaluation across SYNAPS variants: (a) threat prediction accuracy for each model, (b) heatmap of test-set metrics, (c) ROC-AUC curves for each variant, and (d) Precision-Recall curves with average precision for each variant.

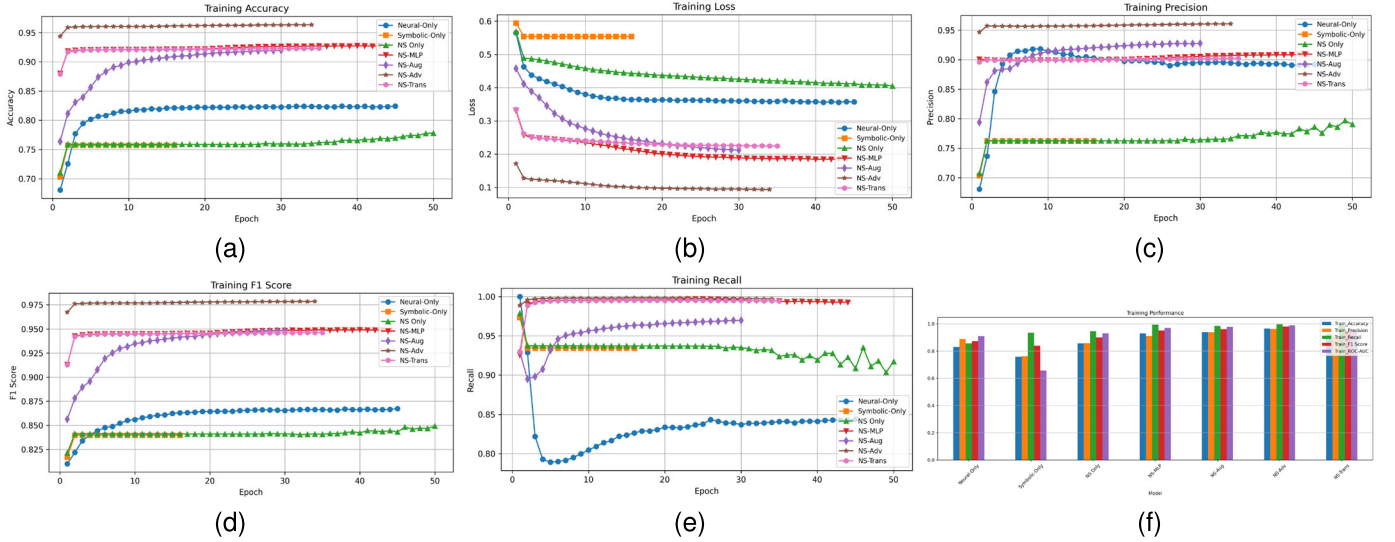


Fig. 5. Training metrics for all SYNAPS variants: (a) training accuracy, (b) training loss, (c) training precision, (d) training F1 score, (e) training recall, and (f) overall training performance comparison across variants.

```

{
  "device_id": "dev_2035",
  "timestamp": "2025-08-01T10:12:44Z",
  "api_calls": ["open_socket", "read_config", "
    send_data"],
  "network_ports": [443],
  "rule_matches": []
}

```

Listing 1: No anomalies and symbolic risk contribution is 0.

$S_{\text{sym}} = 0.0$. Listing 2 illustrates a device report in which several high-severity security rules are triggered. The log

contains suspicious API calls, abnormal port activity, and rule matches that correspond to critical threat patterns. In contrast, the attack example includes unauthorized flash memory access (access_flash, write_flash) of system logs (disable_logging), and use of insecure network ports. Three symbolic rules were fired, with severities of 3, 2, and 3, respectively, with a raw score of $S_{\text{raw}} = (1 \cdot 3) + (1 \cdot 2) + (1 \cdot 3) = 8$. The maximum possible symbolic score for this device type is $S_{\text{max}} = \sum_{i=1}^n \text{severity}_i = 21$. And the normalized symbolic risk score becomes $S_{\text{sym}} = \frac{S_{\text{raw}}}{S_{\text{max}}} = \frac{8}{21} \approx 0.381$ (from equation 17). This non-zero value illustrates that the symbolic module contributes a meaningful and explainable component to the overall risk assessment.

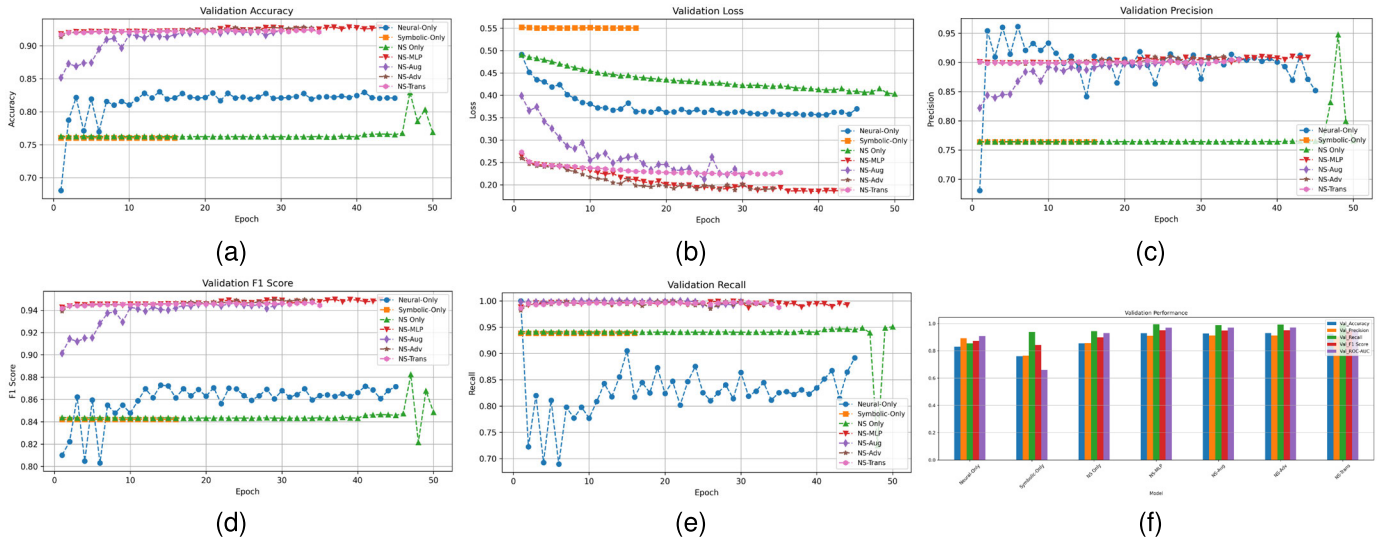


Fig. 6. Validation metrics for all SYNAPS variants: (a) validation accuracy, (b) validation loss, (c) validation precision, (d) validation F1 score, (e) validation recall, and (f) overall validation performance comparison across variants.

```
{
  "device_id": "dev_4591",
  "timestamp": "2025-08-01T14:47:20Z",
  "api_calls": ["access_flash", "write_flash", "
    disable_logging", "open_socket"],
  "network_ports": [23, 8080],
  "rule_matches": [
    {"rule_id": 3, "condition": "
      firmware_write_detected", "severity": 3},
    {"rule_id": 7, "condition": "
      use_of_unsecured_port", "severity": 2},
    {"rule_id": 9, "condition": "
      logging_disabled_during_firmware_access", "
      severity": 3}
  ]
}
```

Listing 2: Multiple high-severity rules fired.

G. Training, Validation, and Test Performance Across Models

Table III reports detailed metrics for each model on training, validation, and test splits. Across all phases, neural-only and symbolic-only models lag behind the hybrid variants. NS-MLP already outperforms both baselines. NS-Adv achieves the best overall results, with a test accuracy of 93.3% and ROC-AUC of 0.971, with a clear benefit from its adversarial robustness training. Fig. 5 and Fig. 6 compare training and validation metrics for all variants. On the other hand, NS-Adv and NS-Aug converge faster and maintain higher precision and recall curves than the other models. The lightweight NS-MLP variant also shows stable learning behavior. The latency and memory consumption results of various SYNAPS variants are summarized in Table IV. Furthermore, the NS-Adv model achieves a balanced trade-off between inference speed and accuracy, with a mean latency of 0.6192 ms and a standard deviation of 0.0545 ms.

H. Future Work and Limitations

The proposed SYNAPS shows strong performance, but it still has some limitations and areas for improvement.

TABLE IV
ON-DEVICE INFERENCE LATENCY AND MEMORY USAGE

Model	Latency (ms)	Std. Dev. (ms)	Memory (MB)
Neural-Only	0.9321	0.0987	1763.27
Symbolic-Only	0.2348	0.0202	1763.27
NS-Only	0.2368	0.0356	1763.27
NS-MLP	0.7331	0.0458	1763.27
NS-Aug	0.6098	0.0261	1763.27
NS-Adv	0.6192	0.0545	1763.27
NS-Trans	0.7040	0.0624	1763.27

At present, the experiments are based mainly on the UNSW-NB15 dataset, which may not capture the full variety of consumer devices or real-world network conditions. The symbolic rules depend on established expert knowledge, which may overlook evolving attack patterns and adaptive adversaries to exploit unrecognized vulnerabilities. The system's real-time efficiency, energy consumption, and uncertainty calibration have not been comprehensively evaluated on resource-constrained devices. In addition, privacy and rigorous threat-model verification need more scrutiny before extensive implementation. In the future, we plan to extend SYNAPS with multi-dataset validation for continuous adaptation and human-in-the-loop rule updates using knowledge graphs. We will add stronger adversarial testing and uncertainty-aware thresholds to improve trust and resilience. To support practical use, future work will also include lightweight model compression for edge devices and federated learning for secure collaboration to ensure consistent protection across device types.

V. CONCLUSION

In this work, we introduced SYNAPS, a hybrid system that combines neural networks and symbolic rules to detect and predict security threats. The proposed proactive approach understands complex patterns and provides explainable decisions. We also improved the system resilience using variant

mechanisms, like data augmentation, adversarial training, and transfer learning, to make it more robust and accurate. Our experiment shows that the advanced versions of SYNAPS perform much better than models that use purely neural or symbolic baselines. The variant models achieved very high accuracy and recall in detecting both common and hidden attacks. Overall, SYNAPS shows that combining neural intelligence with symbolic reasoning can lead to more accurate and explainable security systems for consumer electronic devices. For future work, we plan to extend SYNAPS toward real-time threat monitoring to enable on-the-fly defense against evolving adversarial behaviors.

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